

a correlation between a predictive variable today and next-period returns. In this case, the regression R^2 is simply given by squaring the correlation coefficient, b^2 .

We are also interested in *long-horizon predictability*, which we can measure statistically by extending the regression (8.6) to multiple periods of returns on the left-hand side. We use log returns so we can add them. For example, for a three-period excess return we have

$$r_{t+3} + r_{t+2} + r_{t+1} = c + b \cdot X_t + \varepsilon_{t+3,3}. \quad (8.7)$$

Because we have limited data, we generally run long-horizon regressions using *overlapping* data. That is, when we consider the timing of the variables at time $t+1$ predicting returns over the next three periods, we predict the one-period excess returns from $t+1$ to $t+4$, r_{t+4} , r_{t+3} , and r_{t+2} . Thus, considering the two regressions with right-hand side variables at t and $t+1$, we overlap the one-period returns r_{t+3} and r_{t+2} on the left-hand side. This overlap is behind the complicated notation, $\varepsilon_{t+3,3}$, in the residuals of (8.7), which denotes that the residuals are realized at $t+3$, but they involve returns over the last three periods (the second term in the subscript). The overlapping data turn out to induce some very nasty statistical properties that cause regular ordinary least squares statistical inference to highly overstate how much true predictability there is in data. This is not just a problem for geeky econometricians; it has large implications for investors, as we shall see.

How much predictability should we expect? Not much. Ross (2005, 2012) and Zhou (2010) show that the R^2 in the predictive regression (8.6) should be very low. Technically, Ross and Zhou show that the regression R^2 is bounded by the variance of the pricing kernel (see chapter 6), and the factors entering the pricing kernel, like consumption, are generally not volatile.²² Ross computes a bound of 8% and Zhou's bound is even lower. We should expect R^2 s to be low and typically below 5%. That is, 95% of the movements of the market should be unpredictable, and any strategy that claims to predict future returns with high R^2 s should be viewed with great suspicion. The low degree of predictability is consistent with the near-efficient markets of Grossman and Stiglitz (1980) in chapter 6: profitable market-timing strategies are rare and statistically hard to detect.

The data bear this out. Empirically, it is hard to find robust statistical evidence of predictability. Table 8.7 reports correlation coefficients of next-period returns on various predictors at the beginning of the period. The base unit of measurement is one-quarter, so the one-period ahead return in regression in (8.6) is over the next quarter. I also run long-horizon regressions of the form (8.7) over one year, two years, and five years. These are also reported as correlation coefficients. Table 8.7 reports robust t -statistics that account for the overlapping observation

²² Denoting the pricing kernel as m , Ross (2005) shows that the regression R^2 is bounded by $R^2 \leq (1 + r_f)^2 \text{var}(m)$. Zhou (2010) derives a more restrictive bound of $R^2 \leq \rho_{z,m}^2 (1 + r_f)^2 \text{var}(m)$, where $\rho_{z,m}^2$ is the multiple correlation between the predictive variables z and the pricing kernel m .